# Exploratory Data Analysis on the Automobiles Dataset

Report

## Introduction

The main objective of this Exploratory Data Analysis(EDA) on the Automobiles dataset is to identify patterns, relationships, and key insights concerning sales, mpg, and fuel efficiency of different vehicle models.

Dataset information:

- 205 rows with each vehicle details.
- 26 data columns.
- float64 = 5
- int64 = 5
- object = 16
- Missing values = 205

## Data Cleaning

- Inspect the data set first with the following methods *info()*, *isnull().sum()*
- Remove redundant columns (‘normalized-losses’, ‘symboling’).
- Remove duplicate rows.
- Convert numerical type columns to integer using numpy’s ‘int64’ method.

## Missing Data

- Replace missing data ‘?’ values with ‘*NaN*’.
- Handle missing data properly by removing all ‘NaN’.

# Data Stories and Visualizations

## 1) Hatchback Automobiles

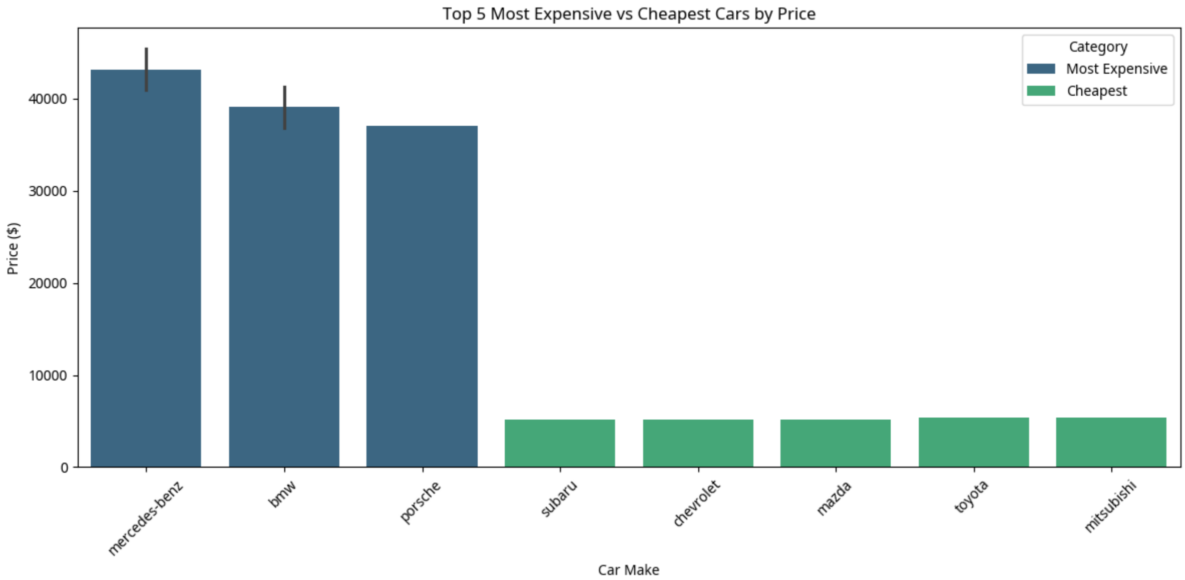
```
Total number of hatchbacks: 70
    symboling  normalized-losses          make  ... city-mpg highway-mpg  price
2           1                122   alfa-romero  ...       19          26  16500
9           0                122          audi  ...       16          22  13207
18          2                121     chevrolet  ...       47          53   5151
19          1                 98     chevrolet  ...       38          43   6295
21          1                118         dodge  ...       37          41   5572
[5 rows x 26 columns]
```

### Method:

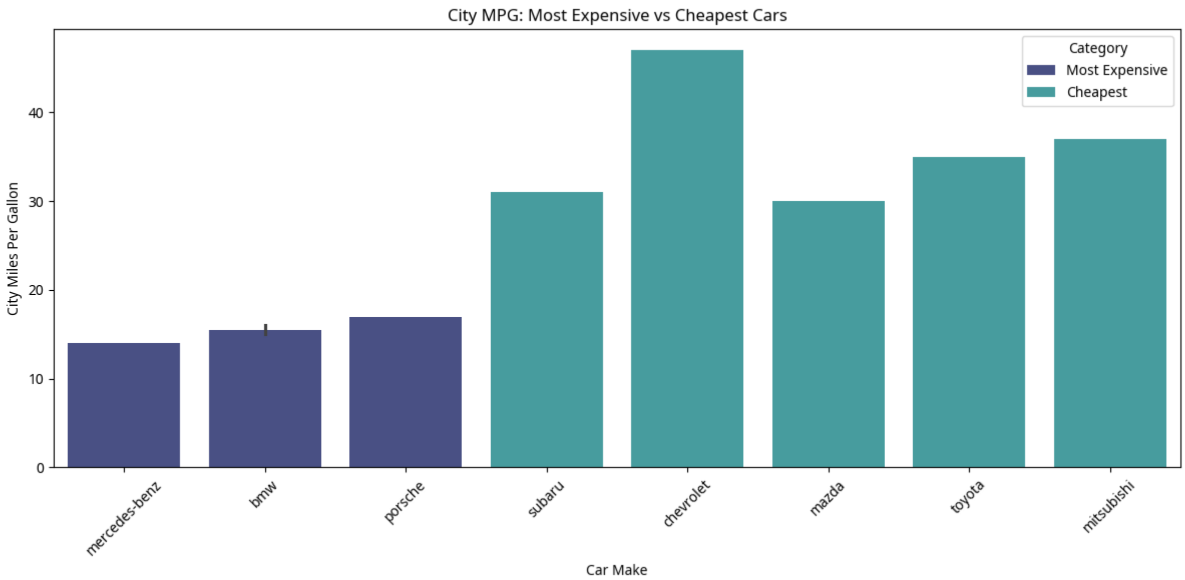
- Create a new variable.
- assign to the dataframe *body-style* column with '*hatchback'* value.
- Use *len()* function to get the length of the new variable.

## 2) Most Expensive vs Cheapest

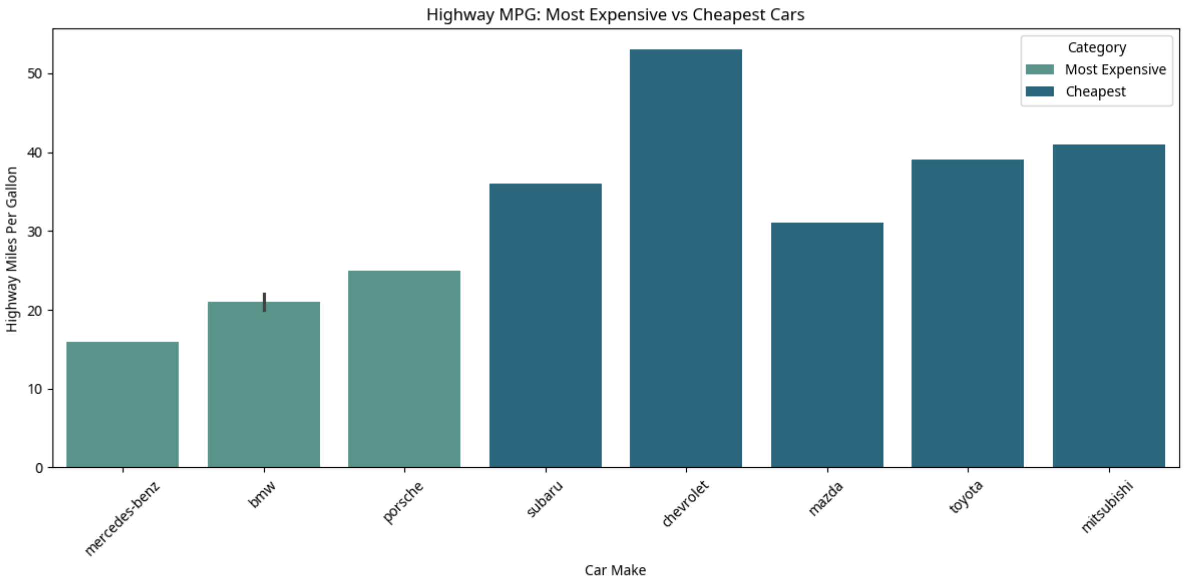
Car price comparison:




City MPG comparison: fuel efficiency on



Highway MPG comparison: fuel efficiency on long drives

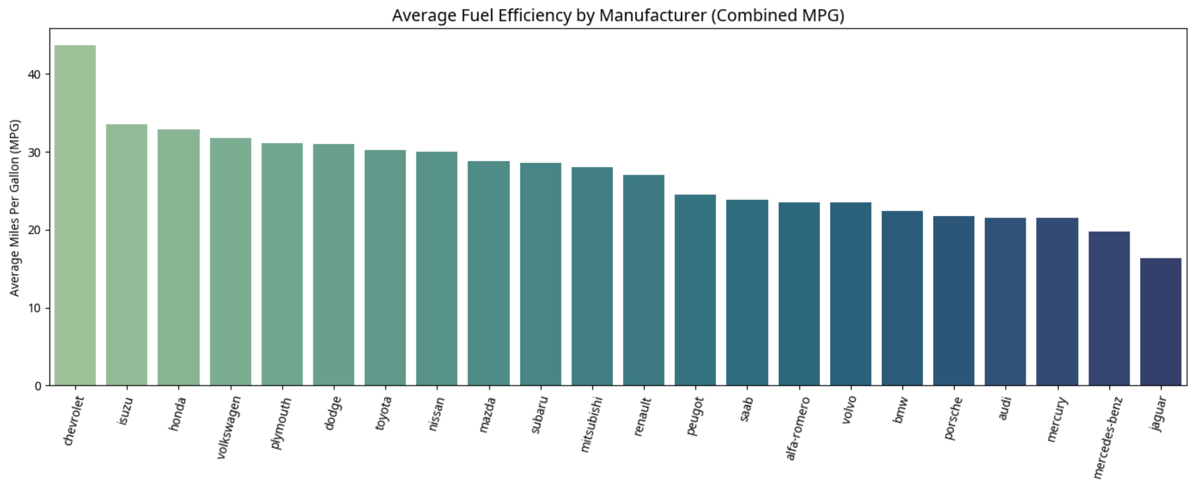



Insights:

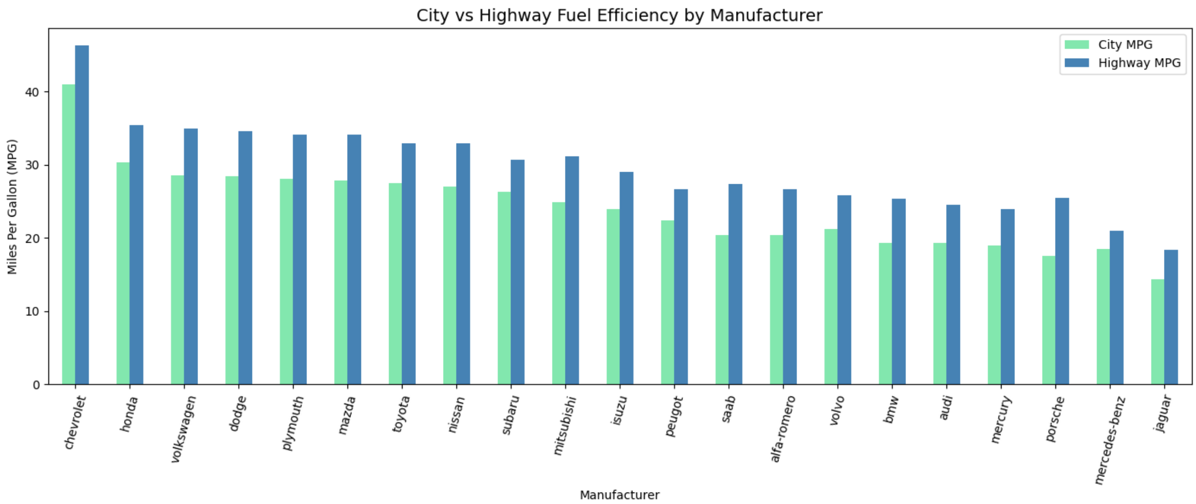
- The most expensive cars tend to have higher prices and lower mpg, emphasizing performance and luxury.
- The cheapest cars offer a higher Mpg which means better fuel economy for budget-conscious consumers.
- Cheaper cars tend to have better city fuel economy.

## 3) Average Fuel Efficiency by Manufacturer (MPG)

Average MPG per Manufacturer, ranking from most to least fuel-efficient:




City MPG vs Highway MPG, brands that perform better on the highway:




Insights:

- Honda, Subaru, Nissan, and Toyota top the chart for fuel efficiency in both city and highway, they are compact and economical cars.
- BMW, Mercedes-Benz, Porsche, and Jaguar tend to appear lower, as they prioritize performance and engine power over fuel economy.
- The gap between city and highway mpg is often larger for sports cars and SUVs.

## 4) Highest Engine Capacity

```
    symboling  normalized-losses           make  ... city-mpg highway-mpg  price
49          0                122         jaguar  ...       13          17  36000
73          0                122  mercedes-benz  ...       14          16  40960
74          1                122  mercedes-benz  ...       14          16  45400
48          0                122         jaguar  ...       15          19  35550
47          0                145         jaguar  ...       15          19  32250
[5 rows x 26 columns]
```

Method:

- Sort the dataset in ascending order based on engine-size column then display first five observations.

Observations:

- Jaguar sedan – engine capacity of 326 cc with 12-cylinders.
- Mercedes-benz – two vehicles have engine size 308 cc with 8-cylinders.
- Porsche hatchback – engine size 203 cc with 6-cylinders.
- BMW – three vehicles have engine size 209 cc with 6-cylinders.

Trends:

Vehicles with the largest engine capacity are primarily luxury or performance brands with body styles like sedans, hardtops, or hatchbacks, and all use fuel.

## 5) Manufacturer with most car models

```
Top 5 manufacturers by number of models:
make
toyota        32
nissan        18
mazda         17
mitsubishi    13
honda         13
Name: count, dtype: int64

Top 5 vehicles with the largest engine sizes:
             make  engine-size body-style fuel-type  price
49         jaguar          326      sedan       gas  36000
73  mercedes-benz          308      sedan       gas  40960
74  mercedes-benz          304    hardtop       gas  45400
48         jaguar          258      sedan       gas  35550
47         jaguar          258      sedan       gas  32250
```

Method:

- Count ‘make’ column for each manufacturer occurrence.

Key Finding:

- Toyota has the most car models, with 32 models.
- Nissan has 18 models, which is significantly lower than first place.
- Mazda with only 17 models.
- Other vehicles have significant representation, but nowhere closer to Toyota.

Insights:

The gap between Toyota and Nissan suggests Toyota has nearly twice as many configurations in the dataset, possibly reflecting its market strategy during the dataset’s time period.

**This report was done by**: Koketso Llale